

苏州大学实验报告

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课程名称	操作系统课程实践					成绩	
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实 验 名 称 实验八 设计一个简单的文件系统

一. 实验目的

- (1) 深入理解 Linux 文件系统的原理。
- (2) 学习并理解 Linux 的 VFS 文件系统管理技术。
- (3) 学习并理解 Linux 的 ext2 文件系统实现技术。
- (4) 设计并实现一个简单的类 ext2 文件系统。

二. 实验内容

设计并实现一个类似于 ext2 但能够对磁盘上的数据块进行加密的文件系统 myext2。本实验的主要内容如下：

- (1) 添加一个类似于 ext2 的文件系统 myext2。
- (2) 修改 myext2 文件系统的 magic number。
- (3) 修改文件系统操作。
- (4) 添加文件系统创建工具。

对于 myext2 文件系统，要求如下：

- (1) myext2 文件系统的物理格式定义与 ext2 文件系统基本一致，但 myext2 文件系统的 magic number 是 0x6666，而 ext2 文件系统的 magic number 是 0xEF53。
- (2) myext2 文件系统是 ext2 文件系统的定制版本，前者

三. 实验步骤

实验环境:

```
fhn@ubuntuserver: ~/OS_experiment/experiment7
fhn@ubuntuserver:~/OS_experiment/experiment7$ neofetch

      .-/+00ssssso+/-.
    `:+ssssssssssssssssss+:`
  -+ssssssssssssssssssyyssss+-
 .ossssssssssssssssssdMMMMyssssso.
 /ssssssssssshdmmNNmyNMMMMhssssss/
 +ssssssssshmydMMMMMMMNddddyssssssss+
 /ssssssssshNMMMyhhyyyhNMMMMNhssssssss/
 .ssssssssdMMMNhssssssssshNMMMdssssssss.
 +ssssshhhyNMMNyssssssssssyyNMMMyssssssss+
 ossyNMMMNyMMhssssssssssshmmhssssssso
 ossyNMMMNyMMhssssssssssshmmhssssssso
 +ssssshhhyNMMNyssssssssssyyNMMMyssssssss+
 .ssssssssdMMMNhssssssssshNMMMdssssssss.
 /ssssssssshNMMMyhhyyyhNMMMMNhssssssss/
 +sssssssssdmydMMMMMMMNddddyssssssss+
 /ssssssssssshdmmNNNmyNMMMMhssssss/
 .ossssssssssssssssssdMMMMyssssso.
 -+ssssssssssssssssssyyssss+-
 `:+ssssssssssssssssss+:`
      .-/+00ssssso+/-.

fhn@ubuntuserver
-----
OS: Ubuntu 22.04.4 LTS aarch64
Host: Parallels ARM Virtual Machine
Kernel: 5.15.0-101-generic
Uptime: 1 hour, 29 mins
Packages: 1835 (dpkg), 12 (snap)
Shell: bash 5.1.16
Resolution: 3024x1890
DE: GNOME 42.9
WM: Mutter
WM Theme: Adwaita
Theme: Yaru [GTK2/3]
Icons: Yaru [GTK2/3]
Terminal: gnome-terminal
CPU: (2)
GPU: 00:0a.0 Red Hat, Inc. Virtio GP
Memory: 1526MiB / 1967MiB
```

(1) 添加一个类似于 ext2 的文件系统 myext2

在 Linux Shell 中执行:

1. `cd /usr/src/linux-source-5.15.0`
2. `cd fs`
3. `cp -R ext2 myext2`
4. `cd /usr/src/linux-source-5.15.0/fs/myext2`
5. `mv ext2.h myext2.h`
6. `cd /lib/modules/$(uname -r)/build/include/linux`
7. `cp ext2_fs.h myext2_fs.h`
8. `cd /lib/modules/$(uname -r)/build/include/asm-generic/bitops`
9. `cp ext2-atomic-setbit.h myext2-atomic-setbit.h`

(2) 修改文件内容

1. 使用如下脚本将原来文件中的 EXT2 替换成 MYEXT2, ext2 替换成 myext2。

1. `#!/bin/bash`
2. `SCRIPT=substitute.sh`

```

3. for f in *; do
4. if [ $f = $SCRIPT ]; then
5. echo "skip $f"
6. continue
7. fi
8. echo -n "substitute ext2 to myext2 in $f..."
9. cat $f | sed 's/ext2/myext2/g' >${f}_tmp
10. echo "done"
11. echo -n "substitute EXT2 to MYEXT2 in $f..."
12. cat $f | sed 's/EXT2/MYEXT2/g' >${f}_tmp
13. mv ${f}_tmp $f
14. echo "done"
15. done

```

脚本命名为 substitute.sh, 在 fs/myext2 目录下执行。注意执行时需要加上可执行权限 (x), 需 删除目录中*.o 文件。

2. 将如下文件

/lib/modules/\$(uname -r)/build/include/linux/myext2_fs.h,
/lib/modules/\$(uname -r)/build/include/asm-generic/bitops/myext2-atomic.h
/lib/modules/\$(uname -r)/build/include/asm-generic/bitops/myext2-atomic-setbit.h
中的 ext2、EXT2 分别替换成 myext2、MYEXT2。

3. 做如下修改:

在 /lib/modules/\$(uname -r)/build/include/asm-generic/bitops.h 文件中添加:

```
#include <asm-generic/bitops/myext2-atomic.h>
```

在 /lib/modules/\$(uname -r)/build/arch/x86/include/asm/bitops.h 文件中添加:

```
#include <asm-generic/bitops/myext2-atomic-setbit.h>
```

在 /lib/modules/\$(uname -r)/build/include/uapi/linux/magic.h 文件中添加:

```
#define MYEXT2_SUPER_MAGIC 0xEF53
```

(3) 把 myext2 编译成内核模块

修改后的 Makefile 文件如下:

```

1. # Makefile for the linux myext2-filesystem routines obj-m := myext2.o
2. myext2-y := balloc.o dir.o file.o ialloc.o inode.o \ ioctl.o namei.o super.o symlink.o
3. KDIR := /lib/modules/$(shell uname -r)/build PWD := $(shell pwd)
4. default:
5. make -C $(KDIR) M=$(PWD) modules

```

在 myext2.h 下增加:

```

1. #include <linux/myext2_fs.h>
2. #include <asm-generic/bitops/myext2-atomic.h>
3. #include <asm-generic/bitops/myext2-atomic-setbit.h>

```

myext2 目录下执行一下指令:

```
1. make # 编译内核模块

2. insmod myext2.ko # 加载文件系统

3. cat /proc/filesystems |grep myext2 # 查看 myext2 文件系统是否加载成功
```

(4) 对 myext2 文件系统进行测试

```
1. cd
2. dd if=/dev/zero of=myfs bs=1M count=1
3. /sbin/mkfs.ext2 myfs
4. mount -t myext2 -o loop ./myfs /mnt
5. mount

6. umount /mnt # 卸载文件系统

7. mount -t ext2 -o loop ./myfs /mnt
8. mount

9. umount /mnt # 卸载文件系统

10. rmmod myext2 # 卸载模块
```

2. 修改 myext2 文件系统的 magic number

找到 myext2 文件系统的 magic number, 并将其值改为 0x6666, 该值在 include/uapi/linux/magic.h, 修改为

```
1. #define MYEXT2_SUPER_MAGIC 0x6666
```

修改后, 使用 make 命令重新编译内核模块, 然后使用 insmod 命令安装编译好的 myext2.ko 内核模块

```
1. make # 编译内核模块

2. insmod myext2.ko # 加载文件系统

3. cat /proc/filesystems |grep myext2 # 查看 myext2 文件系统是否加载成功
```

编写程序 changeMN.c 修改创建的 myfs 文件系统的 magic number, 使其与内核中记录的 myext2 文件系统的 magic number 相匹配:

```

1. #include <stdio.h>
2. int main(int argc, const char **argv) {
3.     int ret;
4.     FILE *fp_read;
5.     FILE *fp_write;
6.     unsigned char buf[2048];
7.     fp_read = fopen(argc >= 2 ? argv[1] : "./myfs", "rb");
8.     if (fp_read == NULL) {
9.         printf("open myfs failed!\n");
10.        return 1;
11.    }
12.    fp_write = fopen(argc >= 3 ? argv[2] : "./fs.new", "wb");
13.    if (fp_write == NULL) {
14.        printf("open fs.new failed!\n");
15.        return 2;
16.    }
17.    ret = fread(buf, sizeof(unsigned char), 2048, fp_read);
18.    printf("previous magic number is 0x%x%x\n", buf[0x438], buf[0x439]); buf[0x438] = 0x66;
19.    buf[0x439] = 0x66;
20.    fwrite(buf, sizeof(unsigned char), 2048, fp_write);
21.    printf("current magic number is 0x%x%x\n", buf[0x438], buf[0x439]); while (ret == 2048)
    {
22.        ret = fread(buf, sizeof(unsigned char), 2048, fp_read); fwrite(buf, sizeof(unsigned
char), ret, fp_write);
23.    }
24.    if (ret < 2048 && feof(fp_read)) { printf("change magic number ok!\n");
25.    }
26.    fclose(fp_read); fclose(fp_write); return 0;
27. }

```

对 changeMN.c 进行编译后，将产生名为 changeMN

```
1. gcc -o changeMN changeMN.c
```

对修改 magic number 后的 myext2 文件系统进行测试：

```

1. dd if=/dev/zero of=myfs bs=1M count=1
2. /sbin/mkfs.ext2 myfs
3. ./changeMN myfs

```

```
4. mount -t myext2 -o loop ./fs.new /mnt
5. mount |tail
6. umount /mnt
7. mount -t ext2 -o loop ./fs.new /mnt
8. rmmod myext2
```

3. 修改文件系统操作

参照 ext2 文件系统中 mknod 操作的实现 ext2_mknod(), 修改 myext2 文件系统下的 mknod 操作:

```
static int myext2_mknod (struct inode * dir, struct dentry *dentry,int mode, int rdev){
    printk(KERN_ERR "oops, `mknod' is not supported by myext2!\n");
    return -EPERM;

//其他代码注释掉
}
```

上述代码指明 mknod 操作不被支持, 并告知 Shell, 将错误号为 EPERM 的结果返回。使用 make 重新编译内核模块, 再使用 insmod 命令安装编译好的 myext2.ko 内核模块:

```
make # 编译内核模块
```

```
insmod myext2.ko # 加载文件系统
```

```
cat /proc/filesystems |grep myext2 # 查看 myext2 文件系统是否加载成功
```

在 Shell 中执行如下测试:

```
mount -t myext2 -o loop ./fs.new /mnt
cd /mnt

mknod myfifo p # 创建一个名为 myfifo 的命名管道

cd

umount /mnt # 卸载文件系统
```

注: 如不显示 printk 打印的信息, 可以通过命令 dmesg|tail 查看。

4.添加文件系统创建工具:

使用如下 Shell 脚本来创建一个 myext2 文件系统创建工具, 命名为 mkfs.myext2:

```
#!/bin/bash

/sbin/losetup -d /dev/loop2
```

```

# 卸载占用的/dev/loop2

/sbin/losetup /dev/loop2 $1

# 第一个参数代表的文件装到/dev/loop2

/sbin/mkfs.ext2 /dev/loop2

# 使用 mkfs.ext2 格式化/dev/loop2

dd if=/dev/loop2 of=./tmpfs bs=1k count=2

# 头 2KB 内容取到临时文件系统

./changeMN $1 ./tmpfs

# 调用程序 changeMN

dd if=./fs.new of=/dev/loop2

# 写回头 2KB 内容

/sbin/losetup -d /dev/loop2

# 卸载文件系统

rm -f ./tmpfs

# 清理临时文件系统

```

脚本的输入是一个文件名，其大小为 myext2 文件系统的大小；脚本的输出则是带有 myext2 文件系统的文件。最后进行如下测试：

```

dd if=/dev/zero of=myfs bs=1M count=1
chmod +x mkfs.myext2

```

```
./mkfs.myext2 myfs
mount -t myext2 -o loop ./myfs /mnt
mount | grep myext2
umount /mnt
rmmod myext2
```

卸载文件系统

卸载模块

四. 实验结果

1. 添加 myext2:

运行 Linux shell 命令的结果:

```
● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ ll
总计 320
drwxr-xr-x 2 root root 4096 5月 9 10:40 ./
drwxr-xr-x 85 root root 4096 5月 9 10:41 ../
-rw-r--r-- 1 root root 6037 5月 9 10:40 acl.c
-rw-r--r-- 1 root root 1550 5月 9 10:40 acl.h
-rw-r--r-- 1 root root 45689 5月 9 10:40 ballocc.c
-rw-r--r-- 1 root root 18523 5月 9 10:40 dir.c
-rw-r--r-- 1 root root 28340 5月 9 10:40 ext2.h
-rw-r--r-- 1 root root 5414 5月 9 10:40 file.c
-rw-r--r-- 1 root root 18818 5月 9 10:40 ialloc.c
-rw-r--r-- 1 root root 49287 5月 9 10:40 inode.c
-rw-r--r-- 1 root root 3840 5月 9 10:40 ioct.c
-rw-r--r-- 1 root root 1257 5月 9 10:40 Kconfig
-rw-r--r-- 1 root root 378 5月 9 10:40 Makefile
-rw-r--r-- 1 root root 10232 5月 9 10:40 namei.c
-rw-r--r-- 1 root root 45323 5月 9 10:40 super.c
-rw-r--r-- 1 root root 846 5月 9 10:40 symlink.c
-rw-r--r-- 1 root root 30056 5月 9 10:40 xattr.c
-rw-r--r-- 1 root root 3367 5月 9 10:40 xattr.h
-rw-r--r-- 1 root root 1511 5月 9 10:40 xattr_security.c
-rw-r--r-- 1 root root 1138 5月 9 10:40 xattr_trusted.c
-rw-r--r-- 1 root root 1273 5月 9 10:40 xattr_user.c
● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ mv ext2.h myext2.h
mv: 无法将 'ext2.h' 移动至 'myext2.h': 权限不够
● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo mv ext2.h myext2.h
● fhn@ubuntu: ~/OS_experiment/experiment8$ cd /lib/modules/$(uname -r)/build/include/linux
fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include$ cp ext2_fs.h myext2_fs.h
cp: 无法创建普通文件 'myext2_fs.h': 权限不够
● fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/linux$ sudo cp ext2_fs.h myext2_fs.h
[sudo] fhn 的密码:
● fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/linux$ cd /lib/modules/$(uname -r)/build/include/asm-generic/bitops
fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/asm-generic/bitops$ cp ext2-atomic.h myext2-atomic.h
cp: 无法创建普通文件 'myext2-atomic.h': 权限不够
● fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/asm-generic/bitops$ sudo cp ext2-atomic.h myext2-atomic.h
● fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/asm-generic/bitops$ sudo cp ext2-atomic-setbit.h myext2-atomic-setbit.h
● fhn@ubuntu: /lib/modules/5.15.0-101-generic/build/include/asm-generic/bitops$
```

(1) 修改文件内容:

运行脚本:

```

● fhn@ubuntu-server: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo bash ./substitute.sh
[sudo] fhn 的密码:
substitute ext2 to myext2 in acl.c...done
substitute EXT2 to MYEXT2 in acl.c...done
substitute ext2 to myext2 in acl.h...done
substitute EXT2 to MYEXT2 in acl.h...done
substitute ext2 to myext2 in balloc.c...done
substitute EXT2 to MYEXT2 in balloc.c...done
substitute ext2 to myext2 in dir.c...done
substitute EXT2 to MYEXT2 in dir.c...done
substitute ext2 to myext2 in file.c...done
substitute EXT2 to MYEXT2 in file.c...done
substitute ext2 to myext2 in ialloc.c...done
substitute EXT2 to MYEXT2 in ialloc.c...done
substitute ext2 to myext2 in inode.c...done
substitute EXT2 to MYEXT2 in inode.c...done
substitute ext2 to myext2 in ioctl.c...done
substitute EXT2 to MYEXT2 in ioctl.c...done
substitute ext2 to myext2 in Kconfig...done
substitute EXT2 to MYEXT2 in Kconfig...done
substitute ext2 to myext2 in Makefile...done
substitute EXT2 to MYEXT2 in Makefile...done
substitute ext2 to myext2 in myext2.h...done
substitute EXT2 to MYEXT2 in myext2.h...done
substitute ext2 to myext2 in namei.c...done
substitute EXT2 to MYEXT2 in namei.c...done
skip substitute.sh
substitute ext2 to myext2 in super.c...done
substitute EXT2 to MYEXT2 in super.c...done
substitute ext2 to myext2 in symlink.c...done
substitute EXT2 to MYEXT2 in symlink.c...done
substitute ext2 to myext2 in xattr.c...done
substitute EXT2 to MYEXT2 in xattr.c...done
substitute ext2 to myext2 in xattr.h...done
substitute EXT2 to MYEXT2 in xattr.h...done
substitute ext2 to myext2 in xattr_security.c...done
substitute EXT2 to MYEXT2 in xattr_security.c...done
substitute ext2 to myext2 in xattr_trusted.c...done
substitute EXT2 to MYEXT2 in xattr_trusted.c...done
substitute ext2 to myext2 in xattr_user.c...done
substitute EXT2 to MYEXT2 in xattr_user.c...done

```

将

/lib/modules/\$(uname -r)/build/include/linux/myext2_fs.h,

/lib/modules/\$(uname -r)/build/include/asm-generic/bitops/myext2-atomic.h

/lib/modules/\$(uname -r)/build/include/asm-generic/bitops/myext2-atomic-setbit.h

中的内容替换:

```
lib > modules > 5.15.0-101-generic > build > include > linux > C myext2_fs.h > ...

1  /* SPDX-License-Identifier: GPL-2.0 */
2  /*
3   * linux/include/linux/ext2_fs.h
4   *
5   * Copyright (C) 1992, 1993, 1994, 1995
6   * Remy Card (card@masi.ibp.fr)
7   * Laboratoire MASI - Institut Blaise Pascal
8   * Universite Pierre et Marie Curie (Paris VI)
9   *
10  * from
11  *
12  * linux/include/linux/minix_fs.h
13  *
14  * Copyright (C) 1991, 1992 Linus Torvalds
15  */
16
17 #ifndef _LINUX_EXT2_FS_H
18 #define _LINUX_EXT2_FS_H
19
20 #include <linux/types.h>
21 #include <linux/magic.h>
22
23 #define EXT2_NAME_LEN 255
24
25 /*
26  * Maximal count of links to a file
27  */
28 #define EXT2_LINK_MAX 32000
29
30 #define EXT2_SB_MAGIC_OFFSET 0x38
31 #define EXT2_SB_BLOCKS_OFFSET 0x04
32 #define EXT2_SB_BSIZE_OFFSET 0x18
33
34 static inline u64 ext2_image_size(void *ext2_sb)
35 {
36     __u8 *p = ext2_sb;
37     if ((__le16 *) (p + EXT2_SB_MAGIC_OFFSET)) != cpu_to_le16(EXT2_SUPER_MAGIC))
```

```
1  /* SPDX-License-Identifier: GPL-2.0 */
2  #ifndef _ASM_GENERIC_BITOPS_myext2_ATOMIC_H
3  #define _ASM_GENERIC_BITOPS_myext2_ATOMIC_H
4
5  /*
6   * Spinlock based version of myext2 atomic bitops
7   */
8
9  #define myext2_set_bit_atomic(lock, nr, addr) \
10  ({ \
11      int ret; \
12      spin_lock(lock); \
13      ret = __test_and_set_bit_le(nr, addr); \
14      spin_unlock(lock); \
15      ret; \
16  })
17
18 #define myext2_clear_bit_atomic(lock, nr, addr) \
19  ({ \
20      int ret; \
21      spin_lock(lock); \
22      ret = __test_and_clear_bit_le(nr, addr); \
23      spin_unlock(lock); \
24      ret; \
25  })
26
27 #endif /* _ASM_GENERIC_BITOPS_myext2_ATOMIC_H */
28
```

```
lib > modules > 5.15.0-101-generic > build > include > asm-generic > bitops > C myext2-atomic-setbit.h > ...
1  /* SPDX-License-Identifier: GPL-2.0 */
2  #ifndef _ASM_GENERIC_BITOPS_EXT2_ATOMIC_SETBIT_H_
3  #define _ASM_GENERIC_BITOPS_EXT2_ATOMIC_SETBIT_H_
4
5  /*
6   * Atomic bitops based version of ext2 atomic bitops
7   */
8
9  #define ext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
10 #define ext2_clear_bit_atomic(l, nr, addr)  test_and_clear_bit_le(nr, addr)
11
12 #endif /* _ASM_GENERIC_BITOPS_EXT2_ATOMIC_SETBIT_H_ */
13
```

在 `/lib/modules/$(uname -r)/build/include/asm-generic/bitops.h` 文件中添加:

```
#include <asm-generic/bitops/myext2-atomic.h>
```

在 `/lib/modules/$(uname -r)/build/arch/x86/include/asm/bitops.h` 文件中添加:

```
#include <asm-generic/bitops/myext2-atomic-setbit.h>
```

在 `/lib/modules/$(uname -r)/build/include/uapi/linux/magic.h` 文件中添加:

```
#define MYEXT2_SUPER_MAGIC 0xEF53
```

```
lib > modules > 5.15.0-101-generic > build > include > asm-generic > C bitops.h > ...
9  /* out before rolling your own buggy implementation in assembly language.
10  */
11  * C language equivalents written by Theodore Ts'o, 9/26/92
12  */
13
14 #include <linux/irqflags.h>
15 #include <linux/compiler.h>
16 #include <asm/barrier.h>
17
18 #include <asm-generic/bitops/ffs.h>
19 #include <asm-generic/bitops/ffz.h>
20 #include <asm-generic/bitops/fls.h>
21 #include <asm-generic/bitops/fls.h>
22 #include <asm-generic/bitops/fls64.h>
23 #include <asm-generic/bitops/find.h>
24
25 #ifndef _LINUX_BITOPS_H
26 #error only <linux/bitops.h> can be included directly
27 #endif
28
29 #include <asm-generic/bitops/sched.h>
30 #include <asm-generic/bitops/ffs.h>
31 #include <asm-generic/bitops/hweight.h>
32 #include <asm-generic/bitops/lock.h>
33
34 #include <asm-generic/bitops/atomic.h>
35 #include <asm-generic/bitops/non-atomic.h>
36 #include <asm-generic/bitops/le.h>
37 #include <asm-generic/bitops/ext2-atomic.h>
38 #include <asm-generic/bitops/myext2-atomic.h>
39
40 #endif /* _ASM_GENERIC_BITOPS_H */
41
```

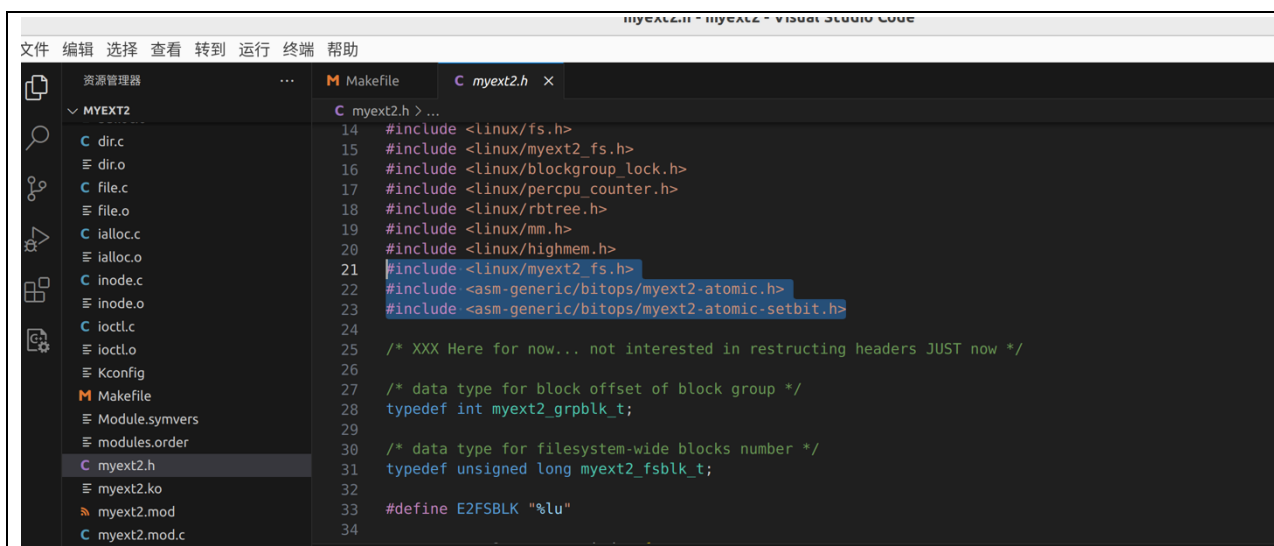
```
lib > modules > 5.15.0-101-generic > build > arch > x86 > include > asm > C bitops.h > ...
11 |
12 | #ifndef _LINUX_BITOPS_H
13 | #error only <linux/bitops.h> can be included directly
14 | #endif
15 |
16 | #include <linux/compiler.h>
17 | #include <asm/alternative.h>
18 | #include <asm/rmwcc.h>
19 | #include <asm/barrier.h>
20 | #include <asm-generic/bitops/myext2-atomic-setbit.h>
21 |
22 | #if BITS_PER_LONG == 32
23 | # define _BITOPS_LONG_SHIFT 5
24 | #elif BITS_PER_LONG == 64
25 | # define _BITOPS_LONG_SHIFT 6
26 | #else
27 | # error "Unexpected BITS_PER_LONG"
```

(2) 把 myext2 编译成内核模块

修改 Makefile:

```
Makefile x
M Makefile
1 # Makefile for the linux myext2-filesystem routines
2 obj-m := myext2.o
3 myext2-y := balloc.o dir.o file.o ialloc.o inode.o \
4 | ioctl.o namei.o super.o symlink.o
5 KDIR := /lib/modules/$(shell uname -r)/build
6 PWD := $(shell pwd)
7 default:
8 | make -C $(KDIR) M=$(PWD) modules
```

修改 myext2.h 中内容:



myext2 目录下执行:

1. make # 编译内核模块
2. insmod myext2.ko # 加载文件系统
3. cat /proc/filesystems |grep myext2 # 查看 myext2 文件系统是否加载成

编译结果:

```
● fh@ubuntu:~/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo make
make -C /lib/modules/5.15.0-101-generic/build M=/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 modules
make[1]: 进入目录 "/usr/src/linux-source-5.15.0-101-generic"
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
  9 | #define myext2_set_bit_atomic(l, nr, addr) test_and_set_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
  9 | #define myext2_set_bit_atomic(lock, nr, addr) \
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
 10 | #define myext2_clear_bit_atomic(l, nr, addr) test_and_clear_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
 18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
     |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
  9 | #define myext2_set_bit_atomic(l, nr, addr) test_and_set_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
  9 | #define myext2_set_bit_atomic(lock, nr, addr) \
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
 10 | #define myext2_clear_bit_atomic(l, nr, addr) test_and_clear_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
 18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
     |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.c:28:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
```

```

        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
 9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
 9 | #define myext2_set_bit_atomic(lock, nr, addr) \
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)  test_and_clear_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
    |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
 9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
 9 | #define myext2_set_bit_atomic(lock, nr, addr) \
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)  test_and_clear_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
    |
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.o
MODPOST /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/Module.symvers
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.mod.o
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
BTF [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
Skipping BTF generation for /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko due to unavailability of vmlinux
make[1]: 离开目录“/usr/src/linux-headers-5.15.0-101-generic”

```

装载模块结果:

```

问题  输出  调试控制台  终端  端口

● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo insmod myext2.ko
○ fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$

```

查看 myext2 是否装载成功:

```

● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ cat /proc/filesystems | grep myext2
myext2
○ fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$

```

2. 修改 myext2 文件系统的 magic number

修改 include/uapi/linux/magic.h:

```
Makefile  myext2.h  magic.h  ●
usr > src > linux-source-5.15.0 > linux-source-5.15.0 > include > uapi > linux > C magic.h > MYEXT2_SUPER_MAGIC
11 #define CRAMFS_MAGIC_WEND 0x43530c2b /* magic number with the wrong endianness */
12 #define DEBUGFS_MAGIC 0x64626720
13 #define SECURITYFS_MAGIC 0x73636673
14 #define SELINUX_MAGIC 0xf97cfff8
15 #define SMACK_MAGIC 0x43415d53 /* "SMAC" */
16 #define RAMFS_MAGIC 0x858458f6 /* some random number */
17 #define TMPFS_MAGIC 0x01021994
18 #define HUGETLBFS_MAGIC 0x958458f6 /* some random number */
19 #define SQUASHFS_MAGIC 0x73717368
20 #define ECRYPTFS_SUPER_MAGIC 0xf15f
21 #define EFS_SUPER_MAGIC 0x414A53
22 #define EROFS_SUPER_MAGIC_V1 0xE0F5E1E2
23 #define EXT2_SUPER_MAGIC 0xEF53
24 #define MYEXT2_SUPER_MAGIC 0x6666
25 #define EXT3_SUPER_MAGIC 0xEF53
26 #define XENFS_SUPER_MAGIC 0xabba1974
27 #define EXT4_SUPER_MAGIC 0xEF53
28 #define BTRFS_SUPER_MAGIC 0x9123683E
29 #define NILFS_SUPER_MAGIC 0x3434
30 #define F2FS_SUPER_MAGIC 0xF2F52010
31 #define HPFS_SUPER_MAGIC 0xf995e849
32 #define TSOS_SUPER_MAGIC 0x0660
```

重新编译内核模块并且安装

文件 编辑 选择 查看 转到 运行 终端 帮助

资源管理器

MYEXT2

- balloc.o.cmd
- dir.o.cmd
- file.o.cmd
- ialloc.o.cmd
- inode.o.cmd
- ioctc.o.cmd
- Module.symvers.cmd
- modules.order.cmd
- myext2.ko.cmd
- myext2.mod.cmd
- myext2.mod.o.cmd
- myext2.o.cmd
- namei.o.cmd
- super.o.cmd
- symlink.o.cmd
- aclic
- aclic
- balloc.c
- balloc.o
- dir.c
- dir.o
- file.c
- file.o
- ialloc.c
- ialloc.o
- inode.c
- inode.o
- ioctc.c
- ioctc.o
- Kconfig
- Makefile
- Module.symvers
- modules.order

问题 输出 调试控制台 终端 端口

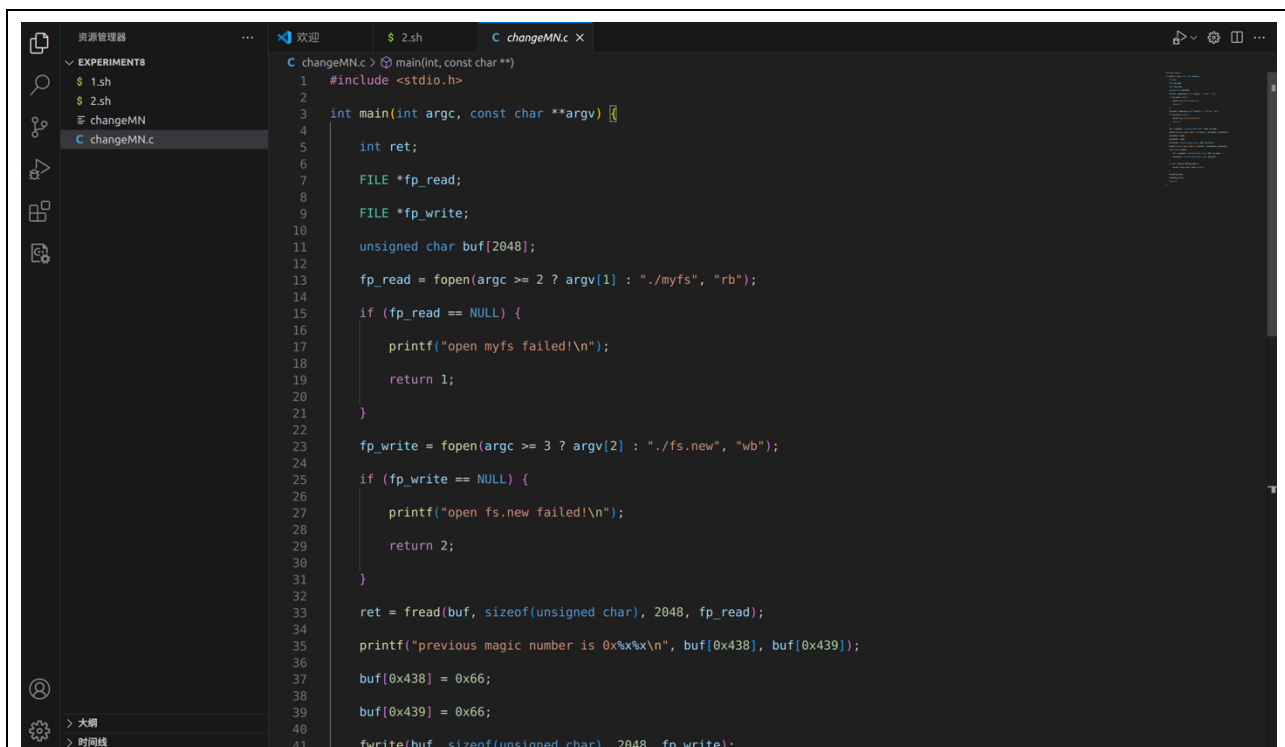
```
● fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo make
[sudo] fhn 的密码:
make -C /lib/modules/5.15.0-101-generic/build M=/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 modules
make[1]: 进入目录 "/usr/src/linux-headers-5.15.0-101-generic"
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
9 | #define myext2_set_bit_atomic(l, nr, addr) test_and_set_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
9 | #define myext2_set_bit_atomic(lock, nr, addr) \
|
In file included from 在编辑器中打开文件 (Ctrl+单击) ./linux-source-5.15.0/fs/myext2/myext2.h:23,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr) test_and_clear_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
|
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
9 | #define myext2_set_bit_atomic(l, nr, addr) test_and_set_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
9 | #define myext2_set_bit_atomic(lock, nr, addr) \
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr) test_and_clear_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
|
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.c:28:
```

```

        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
 9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
 9 | #define myext2_set_bit_atomic(lock, nr, addr)  \
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)  test_and_clear_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
    |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
 9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
 9 | #define myext2_set_bit_atomic(lock, nr, addr)  \
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)  test_and_clear_bit_le(nr, addr)
    |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
        from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
    |
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.o
MODPOST /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/Module.symvers
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.mod.o
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
BTF [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
Skipping BTF generation for /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko due to unavailability of vm
linux
make[1]: 离开目录"/usr/src/linux-headers-5.15.0-101-generic"
fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$
fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ sudo insmod myext2.ko
fhn@ubuntu: /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2$ cat /proc/filesystems | grep myext2
myext2

```

编写程序 changeMN.c 并编译。对修改 magic number 后的 myext2 文件系统进行测试:



```
1 #include <stdio.h>
2
3 int main(int argc, const char **argv) {
4
5     int ret;
6
7     FILE *fp_read;
8
9     FILE *fp_write;
10
11     unsigned char buf[2048];
12
13     fp_read = fopen(argc >= 2 ? argv[1] : "./myfs", "rb");
14
15     if (fp_read == NULL) {
16
17         printf("open myfs failed!\n");
18
19         return 1;
20
21     }
22
23     fp_write = fopen(argc >= 3 ? argv[2] : "./fs.new", "wb");
24
25     if (fp_write == NULL) {
26
27         printf("open fs.new failed!\n");
28
29         return 2;
30
31     }
32
33     ret = fread(buf, sizeof(unsigned char), 2048, fp_read);
34
35     printf("previous magic number is 0x%x\n", buf[0x438], buf[0x439]);
36
37     buf[0x438] = 0x66;
38
39     buf[0x439] = 0x66;
40
41     fwrite(buf, sizeof(unsigned char), 2048, fp_write);
42 }
```

编写 shell 脚本依次运行指令:

```
#!/bin/bash

# 创建一个大小为 1MB 的文件系统镜像

dd if=/dev/zero of=myfs bs=1M count=1

# 格式化为 ext2 文件系统

/sbin/mkfs.ext2 myfs

# 运行自定义命令 changeMN

./changeMN myfs

# 挂载文件系统为 myext2 类型

mount -t myext2 -o loop ./fs.new /mnt

# 显示挂载的文件系统信息

mount | tail

# 卸载挂载的文件系统

umount /mnt

# 挂载文件系统为 ext2 类型

mount -t ext2 -o loop ./fs.new /mnt

# 卸载挂载的文件系统

umount /mnt
```

```
rmmod myext2
```

```
echo "myext2 模块已卸载"
```

shell 脚本运行结果:

```
● fhn@ubuntu:~/OS_experiment/experiment8$ gcc -o changeMN changeMN.c
● fhn@ubuntu:~/OS_experiment/experiment8$ sudo bash 2.sh
[sudo] fhn 的密码:
记录了 1+0 的读入
记录了 1+0 的写出
1048576 字节 (1.0 MB, 1.0 MiB) 已复制, 0.00346693 s, 302 MB/s
mke2fs 1.46.5 (30-Dec-2021)
丢弃设备块: 完成
创建含有 256 个块 (每块 4k) 和 128 个 inode 的文件系统

正在分配组表: 完成
正在写入 inode 表: 完成
写入超级块和文件系统账户统计信息: 已完成

previous magic number is 0x53ef
current magic number is 0x6666
change magic number ok!
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
nsfs on /run/snapd/ns/snap-store.mnt type nsfs (rw)
/var/lib/snapd/snaps/lxd_28384.snap on /snap/lxd/28384 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
gvfsd-fuse on /run/user/1000/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
/var/lib/snapd/snaps/snapd_21467.snap on /snap/snapd/21467 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
/var/lib/snapd/snaps/core22_1383.snap on /snap/core22/1383 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
/var/lib/snapd/snaps/cups_1049.snap on /snap/cups/1049 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
nsfs on /run/snapd/ns/cups.mnt type nsfs (rw)
/var/lib/snapd/snaps/chromium_2847.snap on /snap/chromium/2847 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
nsfs on /run/snapd/ns/chromium.mnt type nsfs (rw)
umount: /mnt: not mounted.
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
umount: /mnt: not mounted.
myext2模块已卸载
● fhn@ubuntu:~/OS_experiment/experiment8$
```

其中

```
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
nsfs on /run/snapd/ns/snap-store.mnt type nsfs (rw)
/var/lib/snapd/snaps/lxd_28384.snap on /snap/lxd/28384 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
gvfsd-fuse on /run/user/1000/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
/var/lib/snapd/snaps/snapd_21467.snap on /snap/snapd/21467 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
/var/lib/snapd/snaps/core22_1383.snap on /snap/core22/1383 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
/var/lib/snapd/snaps/cups_1049.snap on /snap/cups/1049 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
nsfs on /run/snapd/ns/cups.mnt type nsfs (rw)
/var/lib/snapd/snaps/chromium_2847.snap on /snap/chromium/2847 type squashfs (ro,nodev,relatime,errors=continue,x-gdu.hide)
nsfs on /run/snapd/ns/chromium.mnt type nsfs (rw)
umount: /mnt: not mounted.
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
umount: /mnt: not mounted.
```

这一部分是由于为修改了 magic number 的 fs.new 文件无法与 ext2 文件系统匹配

3.修改文件系统操作

修改 myext2_mknod() 函数:

使用 vscode 打开 fs/myext2/namei.c

使用 control + F 找到 myext2_mknod()函数

```

static int myext2_mknod (struct user_namespace * mnt_userns, struct inode * dir,
                        struct dentry *dentry, umode_t mode, dev_t rdev)
{
    struct inode * inode;
    int err;

    err = dqquot_initialize(dir);
    if (err)
        return err;

    inode = myext2_new_inode (dir, mode, &dentry->d_name);
    err = PTR_ERR(inode);
    if (!IS_ERR(inode)) {
        init_special_inode(inode, inode->i_mode, rdev);
        inode->i_op = &myext2_special_inode_operations;
        mark_inode_dirty(inode);
        err = myext2_add_nondir(dentry, inode);
    }
    return err;
}

```

修改函数内容:

```

static int myext2_mknod (struct user_namespace * mnt_userns, struct inode * dir,
                        struct dentry *dentry, umode_t mode, dev_t rdev)
{
    printk(KERN_ERR "haha, mknod is not supported by myext2!you've been cheated!\n");
    return -EPERM;
    // struct inode * inode;
    // int err;

    // err = dqquot_initialize(dir);
    // if (err)
    //     return err;

    // inode = myext2_new_inode (dir, mode, &dentry->d_name);
    // err = PTR_ERR(inode);
    // if (!IS_ERR(inode)) {
    //     init_special_inode(inode, inode->i_mode, rdev);
    //     inode->i_op = &myext2_special_inode_operations;
    //     mark_inode_dirty(inode);
    //     err = myext2_add_nondir(dentry, inode);
    // }
    // return err;
}

```

使用 make 重新编译内核模块, 再使用 insmod 命令安装编译好的 myext2.ko 内核模块:

```

{24-05-14 14:10}ubuntuuser:/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 fh% sudo make
make -C /lib/modules/5.15.0-101-generic/build M=/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 modules
make[1]: 进入目录"/usr/src/linux-headers-5.15.0-101-generic"
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
   9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
   9 | #define myext2_set_bit_atomic(lock, nr, addr) \
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
  10 | #define myext2_clear_bit_atomic(l, nr, addr)    test_and_clear_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/balloc.c:15:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
  18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
     |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
   9 | #define myext2_set_bit_atomic(l, nr, addr)    test_and_set_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
   9 | #define myext2_set_bit_atomic(lock, nr, addr) \
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
  10 | #define myext2_clear_bit_atomic(l, nr, addr)    test_and_clear_bit_le(nr, addr)
     |
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/dir.c:25:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
  18 | #define myext2_clear_bit_atomic(lock, nr, addr) \
     |
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
                 from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/file.c:28:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined

```

```

9 | #define myext2_set_bit_atomic(l, nr, addr)      test_and_set_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
9 | #define myext2_set_bit_atomic(lock, nr, addr)    \
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)     test_and_clear_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/super.c:38:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr)   \
|
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.o
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:9: warning: "myext2_set_bit_atomic" redefined
9 | #define myext2_set_bit_atomic(l, nr, addr)       test_and_set_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:9: note: this is the location of the previous definition
9 | #define myext2_set_bit_atomic(lock, nr, addr)     \
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:23,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic-setbit.h:10: warning: "myext2_clear_bit_atomic" redefined
10 | #define myext2_clear_bit_atomic(l, nr, addr)     test_and_clear_bit_le(nr, addr)
|
In file included from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.h:22,
      from /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/symlink.c:21:
./include/asm-generic/bitops/myext2-atomic.h:18: note: this is the location of the previous definition
18 | #define myext2_clear_bit_atomic(lock, nr, addr)   \
|
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.o
MODPOST /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/Module.symvers
CC [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.mod.o
LD [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
BTF [M] /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko
Skipping BTF generation for /usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2/myext2.ko due to unavailability of vm
linux
make[1]: 离开目录"/usr/src/linux-headers-5.15.0-101-generic"

```

安装模块,并且查看是否安装成功:

```

● {24-05-14 14:12}ubuntuserver:/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 fh% sudo insmod myext2.ko
● {24-05-14 14:13}ubuntuserver:/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 fh% cat /proc/filesystems |grep myex
t2
    myext2
○ {24-05-14 14:13}ubuntuserver:/usr/src/linux-source-5.15.0/linux-source-5.15.0/fs/myext2 fh% █

```

编写 shell 脚本,执行测试程序:

```

#!/bin/bash

# 挂载文件系统为 myext2 类型

mount -t myext2 -o loop ./fs.new /mnt

# 切换到挂载点目录

cd /mnt

# 创建一个名为 myfifo 的命名管道

mknod myfifo p

echo "Named pipe 'myfifo' created in /mnt"

```

```
# 切换回用户的主目录
```

```
cd ~
```

```
# 卸载挂载的文件系统
```

```
umount /mnt
```

```
echo "Filesystem unmounted from /mnt"
```

执行 shell 脚本:

```
myext2模块已卸载
fhn@ubuntu-server:~/05_experiment/experiment8$ sudo bash 3.sh
[sudo] fhn 的密码:
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
Named pipe 'myfifo' created in /mnt
umount: /mnt: not mounted.
Filesystem unmounted from /mnt
```

可以看到 mknod 操作不被支持

4.添加文件系统创建工具

编写 shell 脚本 mkfs.myext2

```
mkfs.myext2 x
mkfs.myext2
1  #!/bin/bash
2
3  /sbin/losetup -d /dev/loop2
4
5  # 卸载占用的/dev/loop2
6
7  /sbin/losetup /dev/loop2 $1
8
9  # 第一个参数代表的文件装到/dev/loop2
10
11 /sbin/mkfs.ext2 /dev/loop2
12
13 # 使用 mkfs.ext2 格式化/dev/loop2
14
15 dd if=/dev/loop2 of=./tmpfs bs=1k count=2
16
17 # 头 2KB 内容取到临时文件系统
18
19 ./changeMN $1 ./tmpfs
20
21 # 调用程序 changeMN
22
23 dd if=./fs.new of=/dev/loop2
24
25 # 写回头 2KB 内容
26
27 /sbin/losetup -d /dev/loop2
28
29 # 卸载文件系统
30
31 rm -f ./tmpfs
32
33 # 清理临时文件系统
```

编写 bash 脚本进行测试:

```
#!/bin/bash

# 创建一个大小为 1MB 的文件系统镜像

dd if=/dev/zero of=myfs bs=1M count=1

# 设置 mkfs.myext2 脚本为可执行

chmod +x mkfs.myext2

# 使用 mkfs.myext2 格式化文件系统镜像

./mkfs.myext2 myfs
```

```
# 挂载文件系统为 myext2 类型

mount -t myext2 -o loop ./myfs /mnt

# 显示挂载的 myext2 文件系统信息

mount | grep myext2

# 卸载挂载的文件系统

umount /mnt

# 卸载 myext2 模块

rmmod myext2

echo "myext2 模块已卸载"
```

执行 bash 脚本:

```
• fhn@ubuntu-server:~/OS_experiment/experiment8$ sudo bash ./4.sh
记录了1+0 的读入
记录了1+0 的写出
1048576字节 (1.0 MB, 1.0 MiB) 已复制, 0.00311403 s, 337 MB/s
losetup: myfs: 设置回环设备失败: 设备或资源忙
mke2fs 1.46.5 (30-Dec-2021)
/dev/loop2 已经挂载; 将不会在此创建文件系统!
记录了2+0 的读入
记录了2+0 的写出
2048字节 (2.0 kB, 2.0 KiB) 已复制, 9.2708e-05 s, 22.1 MB/s
previous magic number is 0x00
current magic number is 0x6666
change magic number ok!
dd: 写入 '/dev/loop2': 不允许的操作
记录了1+0 的读入
记录了0+0 的写出
0字节已复制, 0.00109621 s, 0.0 kB/s
mount: /mnt: wrong fs type, bad option, bad superblock on /dev/loop18, missing codepage or helper program, or other error.
umount: /mnt: not mounted.
myext2 模块已卸载
```

五. 实验总结

在本次实验中, 我通过设计并实现一个简单的类 ext2 文件系统, 深入理解了 Linux 文件系统的工作原理和实现细节。通过对比 ext2 和 myext2 文件系统的实现, 我学会了如何在现有的文件系统基础上进行定制和扩展。实验中遇到的最大挑战是确保所有相关文件名和内容的正确修改, 以避免编译和运行时的错误。

此外, 通过添加文件系统创建工具, 我更加理解了文件系统创建的过程和步骤。未来的工作中, 可以考虑进一步优化 myext2 文件的性能和安全性, 探索更多高级的文件系统功能实现。

总的来说, 本次实验不仅提升了我的理论知识, 还增强了实际动手能力, 为将来的文件系统开发和研究打下了坚实的基础。